

ORBIT: CodeWeaver evidence profile

This profile audits every empirical surface identified for ORBIT and records the maximum scientifically defensible CodeWeaver comparison. Status: measured_existing_slice. All 24 names map exactly to the retained 100-project campaign.

Evidence boundary

Paper values and CodeWeaver measurements remain separate unless subject identity, revision, denominator, and fixed oracle are verified. Missing values are not converted to zeros.

Existing result

No separate compatible CodeWeaver result package.

Complete empirical-surface audit

key	surface	denominator	metrics	status
orbit	Table 1	7 prior/system rows	dataset, programs, projects above 1K LoC, median/mean LoC	exact_reference
orbit	Table 2	24 CRUST-Bench projects	LoC, functions, files, tests	exact_reference
orbit	Table 3	24 CRUST-Bench projects	C2Rust translation/build/test, CRUST-Bench reference build/test, ORBIT expert/generated build/test	exact_reference_and_codeweaver_slice
orbit	Table 4	24 CRUST-Bench projects	pointer dereferences, pointer arithmetic, unsafe LoC, unsafe percent, LoC for four systems	exact_reference
orbit	Table 5	13 TRACTOR programs	performance pass fraction, ORBIT result, vector pass rate	exact_reference
orbit	Table 6	3 CRUST-Bench programs x 4 ablations	function coverage, test coverage, build and test	exact_reference

Inclusion and execution decision

paper_id	source_url	empirical_scope	crust_role	codeweaver_status	reason	existing_result
arXiv:2604.12048	https://arxiv.org/abs/2604.12048	24 named CRUST-Bench projects plus TRACTOR	exact named subset:24	measured_existing_slice	all 24 names map exactly to the retained 100-project campaign	

orbit_table1_scale

system	dataset	programs	percent_over_1kloc	median_loc	mean_loc
RustMap	Rosetta Code, Bzip2	126	~1	~80	~145
Syzygy	Zopfli, URL parser	2	50	2700	2700
EvoC2Rust	C2R-Bench, Vivo-Bench	25	~8	~400	~600
RustAssure	5 C libraries	5	20	405	~900
SmartC2Rust	GitHub, prior studies	21	~24	502	~1000
VERT	TransCoder-IR	534	0	~100	~120
ORBIT	CRUST-Bench	24	91.7	1354	1603

orbit_table4_safety_summary

system	scope	mean_unsafe_percent	zero_unsafe_projects	pointer_declarations	pointer_dereferences	note
C2Rust	15 compiling projects	69.6				range 20.4%-97.4%
CRUST-Bench	11 compiling projects	0.68	8			nonzero: razz_simulation, lambda-calculus-eval, libm17
ORBIT expert	24 projects	0.06	19	35	5	
ORBIT generated	24 projects	0.11	21	58	6	

orbit_table5_tractor

kind	program	performers_passing	performers_total	orbit_result	vector_pass_percent
exec	016_switch-arith	3	6	Pass	100
exec	042_float_union	3	6	Pass	100
exec	033_bitfield	3	6	Pass	100
exec	030_int_underflow	2	6	Pass	100

kind	program	performers_passing	performers_total	orbit_result	vector_pass_percent
exec	002_stdin_echo	3	6	Partial	75
lib	read_scalefactors_lib	3	6	Pass	100
lib	004_loop_lib	2	6	Pass	100
lib	read_side_info_lib	4	6	Pass	100
lib	wcscat_lib	4	6	Pass	100
lib	update_frame_header_lib	4	6	Pass	100
lib	030_int_underflow_lib	2	6	Fail	0
lib	contrast_ratio_lib	3	6	Partial	62.5
lib	hex2bin_lib	2	6	Fail	0

orbit_table6_ablation

project	configuration	function_coverage_percent	test_coverage_percent	build	test
CircularBuffer	base	100	0	True	True
CircularBuffer	without interface	100	90.9	True	True
CircularBuffer	without mapping	100		True	True
CircularBuffer	full	100		True	True
LTRE	base	64.4	3.0	True	True
LTRE	without interface	82.2	12.2	True	True
LTRE	without mapping	93.3		True	True
LTRE	full	100		True	False
libm17	base	84.1	0	False	False
libm17	without interface	100	100	True	True
libm17	without mapping	100		True	True
libm17	full	100		True	True

CodeWeaver exact 24-project summary

system	projects	build_successes	test_successes	test_success_percent	fixed_tests_passed	fixed_tests_expected	fixed_test_rate_percent	protocol
ORBIT expert interfaces	24	24	22	91.6666666667				paper reference; apparent single run

system	projects	build_succe sses	test_succes ses	test_succes s_percent	fixed_tests _passed	fixed_tests _expected	fixed_test_r ate_percent	protocol
ORBIT generated interfaces	24	24	22	91.66666666 667				paper reference; apparent single run
CodeWeav er repetition 1	24	24	11	45.83333333 33333336	97	188	51.5957446 8085106	CodeWeav er multi-stage, 5 repairs, 3 parity rounds
CodeWeav er repetition 2	24	24	11	45.83333333 33333336	95	188	50.5319148 9361702	CodeWeav er multi-stage, 5 repairs, 3 parity rounds
CodeWeav er repetition 3	24	24	11	45.83333333 33333336	79	188	42.0212765 9574468	CodeWeav er multi-stage, 5 repairs, 3 parity rounds